

Formal Notation: The Sieve Filtration Process

The following notation defines the Sieve of Eratosthenes as a recursive set-filtration process. This formalization captures the logistical removal of composite numbers from the candidate set S .

1. Initial State

We define the initial set of candidate integers S_1 up to an upper limit n :

$$S_1 = \{ x \in \mathbb{N} \mid 2 \leq x \leq n \}$$

2. Iterative Filtration Step

At each step k , we identify the next prime p_k as the minimum element of the current set, and perform a subtraction (filtration) of its multiples:

$$p_k = \min(S_k)$$

$$S_{k+1} = S_k \setminus \{ m \cdot p_k \mid m \in \mathbb{N}, m \geq p_k \}$$

Using set-builder notation, the filtration is expressed as:

$$S_{k+1} = \{ x \in S_k \mid x \not\equiv 0 \pmod{p_k} \} \cup \{ p_k \}$$

3. Termination Condition

The filtration process reaches equilibrium when the square of the smallest element in the remaining set exceeds the original limit ***n***:

$$\min(S_k)^2 > n$$

At this condition, the remaining elements in ***S_k*** are inherently prime, and the sieve terminates.